Corporate Bankruptcy Prediction

Machine Learning Mini-Project | UE24CS352A | One-Page Write-up
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1. Problem Statement

Predicting corporate bankruptcy early is valuable to lenders, investors and regulators. This project frames it as a **binary classification** task: given a company's financial attributes, predict whether it is **bankrupt** or **not bankrupt**. We compare four machine-learning classifiers and select the best one using metrics suited to a heavily imbalanced problem.

2. Dataset Details

We use **3year.arff**, the third-year forecasting file of the *Polish Companies Bankruptcy* dataset from the UCI Machine Learning Repository.

Records	10,503 companies	**Features**	64 financial attributes
Bankrupt	495 (about 4.7%)	**Non-bankrupt**	10,008 (about 95.3%)
Data issues	Missing values and duplicate records	**Key challenge**	Severe class imbalance

3. Approach

- **Split:** 80:20 stratified train/test split, preserving the bankrupt/non-bankrupt ratio in both sets.
- **Cleaning:** infinite values treated as missing; missing values imputed with **training-set medians** (computed on training data only, to avoid leakage).
- **Scaling:** features standardized for Logistic Regression and SVM, which are scale-sensitive.
- **Models:** Logistic Regression, RBF-kernel SVM, Random Forest and XGBoost.
- **Evaluation:** accuracy, precision, recall, F1-score, ROC-AUC and PR-AUC. The model with the highest **PR-AUC** is selected, since PR-AUC reflects performance on the rare bankrupt class.

4. Implementation Overview

The project is implemented in a Python notebook that covers data exploration, preprocessing, model training and evaluation, visualizations, feature-importance analysis and sample predictions. In a fresh run on the supplied dataset, **XGBoost achieved the highest PR-AUC (0.6089)** and was selected as the final model.

5. Conclusions

- XGBoost gave the strongest PR-AUC of the four models tested in this run.
- Because bankrupt companies are only about 4.7% of the data, accuracy alone is misleading; precision-recall metrics are more informative.
- The model is intended for academic use and needs further validation before any real-world deployment.